

Assignment 2

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2.1 Equity premium predictors

In this report, we aim to evaluate and compare various approaches to forecast the equity premium, defined as the return on equities in excess of the risk-free rate. Following recent literature, we explore three types of predictors: the historical mean, a fundamental-based decomposition (Ferreira & Santa-Clara, 2011), and a new predictor proposed and validated through our own empirical testing. Our objective is to assess their relative performance in terms of out-of-sample forecasting accuracy.

Using the dataset from data from Goyal, Welch and Zafirov (2022), we implement the return decomposition model of Ferreira & Santa-Clara (2011), which splits expected equity returns into three components, namely the forward dividend yield, the earnings growth and the change in price-to-earnings ratio:

$$E[r_{t+1}] = \left(1 + \frac{D_{t+1}}{P_{t+1}}\right) \cdot \left(\frac{E_{t+1}}{E_t}\right) \cdot \left(\frac{\frac{P_{t+1}}{E_{t+1}}}{\frac{P_t}{E_t}}\right) - 1 \quad (1)$$

To compute the third component, $\frac{P/E_{t+1}}{P/E_t}$, we hypothesized a mean reversion of the P/E ratio over a five-year horizon. We assumed that P/E_{t+5} reverts to the historical average, and we adjusted P/E_{t+1} accordingly using a linear convergence speed of 1/60 per month, since we use monthly data. This more complex implementation is also more robust, as it incorporates long-term equilibrium dynamics rather than relying on short-term fluctuations alone.

We compare the model's forecasts to realized returns using the monthly CRSP series and evaluate performance with out-of-sample RMSE. The RMSE of the sum-of-parts model is 0.08112, while the RMSE of the historical average is 0.05404, indicating that the historical mean performs better in predictive accuracy.

Passing to the new predictor, we employed a stepwise regression approach to improve predictive performance and reduce model complexity. Stepwise regression is a variable selection technique that iteratively adds or removes predictors from the model based on statistical criteria (in our case the p-value), identifying the most informative variables without relying on ad hoc assumptions. This method allows us to isolate a parsimonious set of predictors that enhance out-of-sample forecast accuracy.

The stepwise procedure is particularly useful in our setting, where the signal-to-noise ratio in return predictability is low and the risk of data mining is high due to the large number of potential predictors. To mitigate this, we limited the selection to 12 macro-financial variables widely cited in the literature (e.g., Goyal and Welch, 2004), avoiding arbitrary inputs. Furthermore, we adopted a strict train-test split (0.8–0.2) to ensure that variable selection and performance evaluation were performed on disjoint samples. This procedure substantially reduces the risk of overfitting and helps guard against spurious in-sample relationships.

The final model, selected on the training set, included earnings-to-price (e/p) and the variance risk premium (vrp) as the most relevant predictors. The resulting predictive regression takes the following form:

$$r_{t+1} = \alpha + \beta_1(e/p)_t + \beta_2(vrp)_t + \varepsilon_{t+1} \quad (2)$$

When evaluated on the test set, this model achieved an out-of-sample RMSE of 0.0607. Both coefficients were positive and statistically significant (p -values < 0.01). These results confirm the model's validity and highlight its ability to outperform the sum-of-parts benchmark while maintaining economic interpretability. The fact that the two selected predictors are supported by theory and previous empirical work (e.g., Bollerslev, Tauchen, and Zhou 2009) further strengthens the credibility of the model.

In conclusion, although the stepwise model does not outperform the historical mean in pure predictive accuracy, it offers a more informative and theoretically grounded framework for understanding equity premium variation. This structured approach to variable selection ensures robustness against data-mining concerns and supports more meaningful economic interpretation than either naïve averaging or unrestricted multivariate models.

2.2 Trading strategy based on equity premium indicators

In this section, we develop a dynamic long-short trading strategy that allocates capital between equity and 10-year U.S. Treasury bond. The aim is to maximize expected returns while managing risk through a mean-variance optimization framework. Based on the previous forecasting analysis, we use the historical mean as the best-performing predictor.

Expected equity returns are estimated using a rolling historical mean, requiring at least 20 years of data to ensure stability. For bonds, expected returns are approximated by the prevailing 10-year Treasury yield, adjusted for monthly frequency. Asset weights are determined each month by solving a mean-variance optimization problem that balances expected return against risk, subject to two constraints: the strategy is cash-neutral, meaning the net investment

across both assets is zero, and it is unleveraged, meaning the total exposure across long and short positions does not exceed 100%.

To model risk, we assume equity and bond returns are uncorrelated and estimate their volatilities using an exponentially weighted moving average (EWMA) method. This technique gives more weight to recent observations and allows the model to respond quickly to changing market conditions. We use a three-year half-life for the volatility estimates, providing a balance between responsiveness and stability. Figure 1 shows the cumulative return of the long-short strategy over the full sample period, beginning with a \$1 investment. The strategy demonstrates strong performance, with steady compounding growth and relatively limited drawdowns. It navigates major market disruptions, including the dot-com crash, the 2008 financial crisis, and the COVID-19 shock, while continuing to deliver positive returns. Overall, the strategy delivers a moderate level of volatility and a positive Sharpe ratio, confirming its effectiveness in generating risk-adjusted returns over the long term (Table 1).

To evaluate the benefit of risk modeling, we compare two versions of the strategy: Risk Model ON, which adjusts weights based on expected returns and volatility, and Risk Model OFF, a naive version based solely on expected returns. In the naive strategy, equity weights are fixed at ± 0.5 (due to constraints) depending on the return signal, with no adjustment for risk. This leads to constant exposure, even during turbulent periods. By contrast, the risk-aware strategy reduces exposure when expected volatility rises.

To illustrate this, we compared the equity weights assigned by each strategy during the 2008–2009 crisis period (Figure 2). The results show a clear divergence: the risk-aware portfolio cut equity exposure sharply during the peak of market stress, while the naive portfolio held constant weights. This comparison highlights the benefits of incorporating a dynamic risk model, namely, improved drawdown management and more stable performance during periods of market turbulence.

	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio
Strategy Performance	5.13	6.40	0.80

Table 1: Risk-Adjusted Performance of the Forecast-Based Strategy

2.3 Trading strategy performance

In this analysis, we examine the performance of the long-short trading strategy developed in the previous paragraph by running a time-series regression of the performance (using the historical mean return) on various known factors, following the methodology of *Frazzini et al. (2013)*. The data collected includes monthly returns on US (or global, if US data was not available) factors: Market Risk (Mkt - RF); Size (SMB - Small Minus Big); Value (HML - High Minus Low); Profitability (RMW - Robust Minus Weak); Investment (CMA - Conservative Minus Aggressive); Quality Minus Junk (QMJ); Momentum (MOM); Equity Carry (Carry - EQ); Fixed Income Carry (Carry - FI). The goal is to determine the alpha and betas of the strategy, assess its economic and statistical significance and identify the specific factor exposures driving the strategy’s performance. The results are shown in the Appendix (Figure 3).

The alpha of the strategy is 0.0011, which is statistically significant with a t-value of 5.424 and a p-value less than 0.0001. Given the context of financial returns, the magnitude of the alpha is considered economically large, especially when compounded over time. Considering a 5% confidence level, the performance of the strategy is driven by both factor exposures and alpha. The significant alpha might suggest skill or unique strategy characteristics, given that the strategy generates a positive return that is not explained by the included factors. The significant coefficients for Market Risk (0.0044, $p < 0.0001$), Size (-0.0008, $p < 0.0001$), and Fixed Income Carry (0.0002, $p < 0.0001$) indicate that these factors also play a role in driving the strategy’s performance. These exposures are statistically significant and suggest that the strategy benefits from the overall market movements and fixed income carry, while being negatively impacted by size.

To assess the robustness of the market exposures, we conducted a rolling window analysis of the betas for all factors and the alpha over 36-month rolling windows. The rolling beta plots for each factor and the rolling alpha provide insights into how the strategy’s sensitivities to these factors have evolved over time. The rolling beta for Mkt-RF (Market Risk) shows that the strategy’s exposure to market risk has remained consistently positive, with some fluctuations. The rolling beta for SMB (Size) reveals a consistent negative exposure, indicating a robust and persistent preference for larger-cap stocks over smaller-cap stocks. The rolling betas for Fixed Income Carry shows little fluctuation, reflecting the strategy’s responsiveness to interest rate environments and yield differentials. The rolling alpha plot shows that the strategy’s alpha has remained consistently positive, demonstrating its robustness and effectiveness over time. Figure 4 in the appendix shows the rolling betas of the statistically significant factors and the rolling alpha. The rolling betas for all other factors show some variability. Overall, the analysis highlights the dynamic nature of the strategy’s factor exposures and its ability to adapt to changing market conditions while consistently generating positive alpha. This adaptability and robustness are key strengths of the strategy, allowing it to navigate different market environments effectively.

Appendix

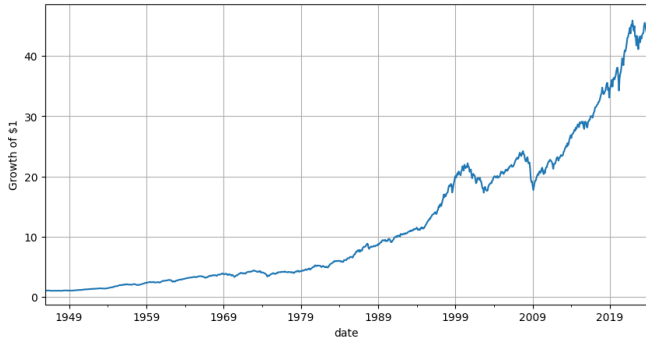


Figure 1: Cumulative return for long-short strategy

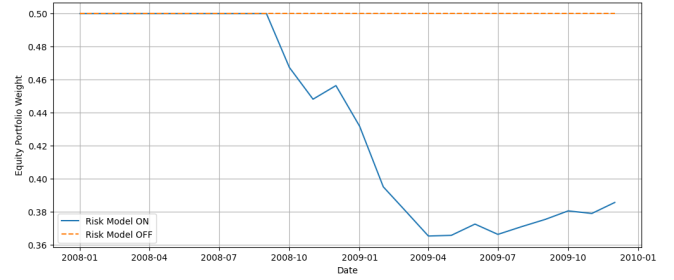


Figure 2: Equity weights during global financial crisis (2008-2009)

OLS Regression Results						
Dep. Variable:	Strategy_Returns		R-squared:	0.980		
Model:	OLS		Adj. R-squared:	0.980		
Method:	Least Squares		F-statistic:	1499.		
Date:	Mon, 21 Apr 2025		Prob (F-statistic):	3.47e-224		
Time:	19:05:04		Log-likelihood:	1257.3		
No. Observations:	279		AIC:	-2495.		
Df Residuals:	269		BIC:	-2458.		
Df Model:	9					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0011	0.000	5.424	0.000	0.001	0.001
MOM	0.0003	0.000	0.037	0.970	-0.017	0.017
Mkt-RF	0.0044	5.83e-05	76.282	0.000	0.004	0.005
SMB	-0.0008	6.06e-05	-13.798	0.000	-0.001	-0.001
HML	0.0002	8.56e-05	1.899	0.059	-5.97e-06	0.000
RMW	0.0001	0.000	1.358	0.175	-6.15e-05	0.000
CMA	-0.0001	0.000	-1.304	0.193	-0.000	7.47e-05
QMJ	0.0273	0.016	1.743	0.082	-0.004	0.058
Carry - EQ	-2.631e-05	3.56e-05	-0.740	0.460	-9.63e-05	4.37e-05
Carry - FI	0.0002	6.23e-05	3.650	0.000	0.000	0.000
Omnibus:	3.706		Durbin-Watson:	1.508		
Prob(Omnibus):	0.157		Jarque-Bera (JB):	3.670		
Skew:	-0.170		Prob(JB):	0.160		
Kurtosis:	3.447		Cond. No.	558.		

Figure 3: Regression output

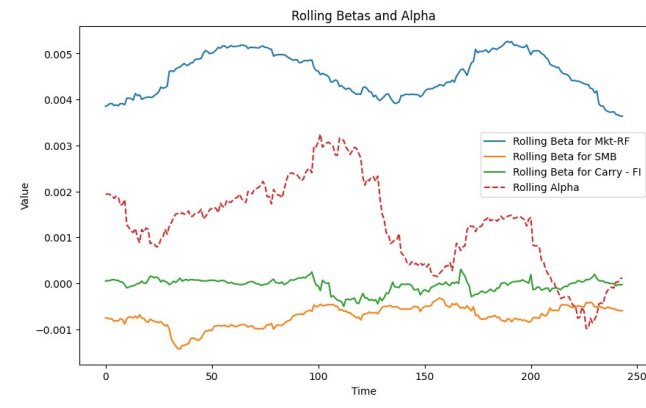


Figure 4: Rolling Betas and Alphas

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